

Hardware Inventory - Data Dictionary

Data Dictionary

Product: Hardware Inventory System

Source of truth: [prisma/schema.prisma](#)

Version: 1.0

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Related docs: [ER_DIAGRAMS.md](#) (visual ER), [FSD.md](#) §8, [FRD.md](#)

This document describes every persisted entity, field, enum, and foreign key in the PostgreSQL schema. **ER diagrams** live in a separate file: [ER_DIAGRAMS.md](#).

1. Conventions

Convention	Meaning
PK	Primary key
FK	Foreign key
UQ	Unique constraint (column or composite) — shown as UK in Mermaid ER attribute blocks
?	Nullable
Decimal(p,s)	Prisma @db.Decimal(p, s)
cuid()	Client-generated CUID string id
Money	Amounts typically Decimal(14,2); unit costs/prices often Decimal(14,4); stock qty Decimal(18,4)
Timestamps	createdAt default now(); updatedAt auto on update where present

Cascade notes are called out per FK (e.g. deleting a product cascades barcodes/units; deleting a sale cascades lines/payments).

2. Entity overview

Entity	Purpose
User	Staff account (login, role, audit actor)
Supplier	Vendor master for purchases
Customer	Buyer master (optional link on sales)
Product	Catalog item + on-hand stock + pricing/GST
ProductUnit	Sell/buy unit with conversion to base
Barcode	Scan code on product or unit
Promotion / PromotionProduct	Optional %-off rules (UI may be flagged off)
Purchase / PurchaseLine / PurchasePayment	Stock-in + supplier payables
Sale / SaleLine / SalePayment	Stock-out + customer receivables
Quotation / QuotationLine / QuotationCounter	Customer quotation; does not move stock
StockMovement	Quantity ledger (typed in/out)
StockAdjustment	Explicit correction audit
ReportScheduleConfig	Singleton owner-report schedule
ReportRun	Saved scheduled report instance

See also: [ER_DIAGRAMS.md](#) for relationship and domain diagrams.

3. Enumerations

UserRole

Value	Meaning
ADMIN	Full back-office access
MANAGER	Treated as admin (legacy compat)
CASHIER	Counter / read-limited access

ProductStatus

Value	Meaning
ACTIVE	Sellable / listed
INACTIVE	Soft-disabled

UnitKind

Value	Meaning
PIECE	Discrete count
WEIGHT	Mass
LENGTH	Length
VOLUME	Volume
PACK	Pack/case style
OTHER	Unclassified

BarcodeType

Value	Meaning
PRODUCT	Product-level code
PACK	Pack barcode
INTERNAL	Internal shop code
SUPPLIER	Supplier barcode

PurchaseStatus

Value	Meaning
DRAFT	Not finalized
RECEIVED	Stock received / posted
CANCELLED	Cancelled

SaleStatus

Value	Meaning
DRAFT	Open / incomplete
COMPLETED	Posted sale
CANCELLED	Cancelled
RETURNED	Return state

QuotationStatus

Stored on Quotation. **Expired is not a stored value.** An ISSUED quotation is treated as expired when the current Asia/Kolkata date is after `validUntil` (the valid-until day itself is still valid).

Value	Meaning
DRAFT	Editable. Not a final quotation
ISSUED	Final. Pricing is frozen. Printable

Value	Meaning
CANCELLED	Cancelled by an admin. Kept for history

SalePaymentMethod

Value	Meaning
CASH	Cash tender
ONLINE_BANKING	Online / UPI / bank transfer style

PromotionScope

Value	Meaning
CART	Whole cart
PRODUCT	Specific products (via link table)
CATEGORY	Category string match

StockMovementType

Value	Direction (typical)
OPENING	Opening balance in
PURCHASE_IN	Purchase receipt
SALE_OUT	Sale deduction
SALE_RETURN_IN	Sale return
PURCHASE_RETURN_OUT	Purchase return
ADJUSTMENT_IN / ADJUSTMENT_OUT	Manual adjust
DAMAGE_OUT / WASTAGE_OUT	Loss
RECOUNT_IN / RECOUNT_OUT	Recount

ScheduledReportType

Value	Meaning
SALES	Sales + margin period report
INVENTORY	Stock snapshot
SUPPLIER_OUTSTANDING	Unpaid purchases
CUSTOMER_OUTSTANDING	Unpaid sales

ReportPeriod

Value	Meaning
DAILY	Previous calendar day (IST rules in services)

Value	Meaning
MONTHLY	Previous month
QUARTERLY	Previous Indian FY quarter
YEARLY	Previous year window

ReportRunStatus

Value	Meaning
PENDING	Queued
RUNNING	In progress
COMPLETED	Success (payload stored)
FAILED	Error (errorMessage)

4. Table / field dictionary

4.1 User

Staff identity for login and audit.

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	User id
fullName	String	N			Display name
username	String	N		UK	Login name
email	String	Y		UK	Optional email
phone	String	Y		UK	Optional phone
passwordHash	String	N			bcrypt hash (never plaintext)
role	UserRole	N	CASHIER		Access role
isActive	Boolean	N	true		Inactive users cannot auth
createdAt	DateTime	N	now()		Created
updatedAt	DateTime	N	auto		Updated

Indexes: (role, isActive)

Relations: created purchases/sales; stock movements; stock adjustments; sale/purchase payments.

4.2 Supplier

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Supplier id

Field	Type	Null	Default	Keys	Description
name	String	N		UK	Supplier name
contactPerson	String	Y			Contact name
phone	String	Y		UK	Phone
email	String	Y			Email
address	String	Y			Address
gstNumber	String	Y			GSTIN
note	String	Y			Notes
createdAt	DateTime	N	now()		Created
updatedAt	DateTime	N	auto		Updated

Relations: purchases → Purchase

4.3 Customer

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Customer id
name	String	N			Name
phone	String	Y		UK	Phone
email	String	Y			Email
address	String	Y			Address
partyGstNo	String	Y			Party GST for invoices
partyState	String	Y			State / place for tax invoice
createdAt	DateTime	N	now()		Created
updatedAt	DateTime	N	auto		Updated

Indexes: name

Relations: sales → Sale (onDelete: SetNull on sale.customerId)

4.4 Product

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Product id
sku	String	N		UK	Stock keeping unit
name	String	N			Product name
slug	String	Y		UK	Optional URL-ish slug
description	String	Y			Description
category	String	Y			Category label
brand	String	Y			Brand name

Field	Type	Null	Default	Keys	Description
brandCode	String	Y		UK*	Business code; *case-insensitive unique when non-blank (DB index)
color	String	Y			Color
size	String	Y			Size/dimension label
hsnCode	String	Y			HSN for GST
status	ProductStatus	N	ACTIVE		Active/inactive
baseUnitCode	String	N			Code of base unit
unitKind	UnitKind	N			Kind of measure
allowsFractional	Boolean	N	false		Fractional base qty allowed
costPrice	Decimal(14,4)	Y			Latest / reference cost
avgCostPrice	Decimal(14,4)	Y			Moving weighted average cost
percentage	Decimal(8,4)	Y			Markup %
mrp	Decimal(14,4)	Y			MRP
sellingPrice	Decimal(14,4)	Y			Shelf selling price
cgstPercent	Decimal(5,2)	Y			CGST %
sgstPercent	Decimal(5,2)	Y			SGST %
igstPercent	Decimal(5,2)	Y			IGST %
priceReviewNeeded	Boolean	N	false		Cheaper purchase → review flag
suggestedSellingPrice	Decimal(14,4)	Y			Suggested markdown
priceReviewNote	String	Y			Review reason text
reorderLevel	Decimal(14,4)	Y			Low-stock threshold (base units)
currentStock	Decimal(18,4)	N	0		On-hand in base units
createdAt	DateTime	N	now()		Created
updatedAt	DateTime	N	auto		Updated

Indexes: name, category, brand, status

4.5 ProductUnit

Field	Type	Null	Default	Keys	Descrip
id	String	N	cuid()	PK	Unit id
productId	String	N		FK → Product	Parent product (onDelete Cascade)
code	String	N		UK(productId,code)	Unit code
displayName	String	N			Display label
isBaseUnit	Boolean	N	false		True for base unit row
conversionToBase	Decimal(18,4)	N			Multiply qty → base qty
allowsFractionalSale	Boolean	N	false		Fractional sale in this unit
createdAt	DateTime	N	now()		Created
updatedAt	DateTime	N	auto		Updated

Indexes: (productId, isBaseUnit)

4.6 Barcode

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Barcode row id
code	String	N		UK	Scanned value
type	BarcodeType	N	PRODUCT		Barcode kind
productId	String	N		FK → Product	Product (Cascade)
productUnitId	String	Y		FK → ProductUnit	Optional unit (SetNull)
note	String	Y			Note
createdAt	DateTime	N	now()		Created
updatedAt	DateTime	N	auto		Updated

4.7 Promotion

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Promotion id
name	String	N			Name
code	String	Y		UK	Optional promo code
scope	PromotionScope	N			CART / PRODUCT / CATEGORY
category	String	Y			Category when scope=CATEGORY
percentage	Decimal(5,2)	N			Discount %
isActive	Boolean	N	true		Active flag
startsAt	DateTime	Y			Window start
endsAt	DateTime	Y			Window end
note	String	Y			Note
createdAt	DateTime	N	now()		Created
updatedAt	DateTime	N	auto		Updated

Indexes: (scope, isActive), category

4.8 PromotionProduct

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Link id
promotionId	String	N		FK → Promotion	(Cascade)
productId	String	N		FK → Product	(Cascade)
createdAt	DateTime	N	now()		Created

Unique: (promotionId, productId)

4.9 Purchase

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Purchase id
purchaseNumber	String	N		UK	Human-readable number
supplierId	String	N		FK → Supplier	Supplier
status	PurchaseStatus	N	DRAFT		Lifecycle
invoiceNumber	String	Y			Supplier invoice #

Field	Type	Null	Default	Keys	Description
invoiceDate	DateTime	Y			Supplier invoice date
note	String	Y			Note
subtotal	Decimal(14,2)	N	0		Lines subtotal
discountAmount	Decimal(14,2)	N	0		Header discount
taxAmount	Decimal(14,2)	N	0		Tax
totalAmount	Decimal(14,2)	N	0		Total
paidAmount	Decimal(14,2)	N	0		Sum of payments
balanceAmount	Decimal(14,2)	N	0		Amount still owed
createdById	String	N		FK → User	Creator
createdAt	DateTime	N	now()		Created
updatedAt	DateTime	N	auto		Updated

Indexes: (supplierId, status), invoiceDate

Relations: lines, payments, stockMovements

4.10 PurchaseLine

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Line id
purchaseId	String	N		FK → Purchase	Parent (Cascade)
productId	String	N		FK → Product	Product
productUnitId	String	N		FK → ProductUnit	Unit purchased
quantity	Decimal(18,4)	N			Qty in selected unit
quantityInBase	Decimal(18,4)	N			Qty in base units
unitCost	Decimal(14,4)	N			Cost per selected unit
lineDiscount	Decimal(14,2)	N	0		Line discount
lineTax	Decimal(14,2)	N	0		Line tax
lineTotal	Decimal(14,2)	N			Line total
createdAt	DateTime	N	now()		Created
updatedAt	DateTime	N	auto		Updated

4.11 PurchasePayment

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Payment id
purchaseId	String	N		FK → Purchase	Parent (Cascade)
amount	Decimal(14,2)	N			Amount paid
paidAt	DateTime	N	now()		Payment date (may be backdated)
note	String	Y			Note
createdById	String	N		FK → User	Who recorded
createdAt	DateTime	N	now()		Row created

Indexes: (purchaseId, paidAt), createdById

4.12 Sale

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Sale id
saleNumber	String	N		UK	Bill/tax series number
status	SaleStatus	N	DRAFT		Lifecycle
customerName	String	Y			Walk-in / entered name
customerPhone	String	Y			Entered phone
customerId	String	Y		FK → Customer	Linked customer (SetNull)
customerNameSnapshot	String	Y			Name snapshot at sale
customerPartyGstNo	String	Y			GST snapshot for invoice
customerPartyState	String	Y			State snapshot for invoice
note	String	Y			Note
subtotal	Decimal(14,2)	N	0		Lines subtotal
discountAmount	Decimal(14,2)	N	0		Header discount

Field	Type	Null	Default	Keys	Description
taxAmount	Decimal(14,2)	N	0		Tax
transportAmount	Decimal(14,2)	N	0		Freight (not per line)
totalAmount	Decimal(14,2)	N	0		Charged total
paidAmount	Decimal(14,2)	N	0		Sum of payments
balanceAmount	Decimal(14,2)	N	0		Outstanding balance
createdById	String	N		FK → User	Creator
createdAt	DateTime	N	now()		Created
updatedAt	DateTime	N	auto		Updated

Indexes: (status, createdAt), customerPhone, customerId

4.13 SaleLine

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Line id
saleId	String	N		FK → Sale	Parent (Cascade)
productId	String	N		FK → Product	Product
productUnitId	String	N		FK → ProductUnit	Unit sold
quantity	Decimal(18,4)	N			Qty in selected unit
quantityInBase	Decimal(18,4)	N			Qty in base units
unitPrice	Decimal(14,4)	N			Price per selected unit
lineDiscount	Decimal(14,2)	N	0		Line discount
lineTax	Decimal(14,2)	N	0		Line tax
lineTotal	Decimal(14,2)	N			Line total
createdAt	DateTime	N	now()		Created
updatedAt	DateTime	N	auto		Updated

4.14 SalePayment

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Payment id
saleId	String	N		FK → Sale	Parent (Cascade)

Field	Type	Null	Default	Keys	Description
amount	Decimal(14,2)	N			Amount
method	SalePaymentMethod	N	CASH		Cash or online banking
note	String	Y			Note
createdById	String	N		FK → User	Who recorded
createdAt	DateTime	N	now()		Created

Indexes: (saleId, createdAt)

4.14a QuotationCounter

One row per calendar year. The create transaction locks this row so QUO-YYYY-#### cannot collide.

Field	Type	Null	Default	Keys	Description
year	Int	N		PK	Asia/Kolkata year
lastNumber	Int	N	0		Last allocated sequence

4.14b Quotation

Does not post stock. convertedSaleId is reserved for a future sale created through /api/sales.

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Quotation id
quotationNumber	String	N		UK	QUO-YYYY-####
customerId	String	Y		FK → Customer	Optional link (SetNull)
status	QuotationStatus	N	DRAFT		Stored status. Expired is derived
quotationDate	Date	N			India calendar date
validUntil	Date	N			Quotation date + 2 days
customerName	String	N			Snapshot

Field	Type	Null	Default	Keys	Description
customerContactPerson	String	Y			Snapshot only
customerPhone	String	Y			Snapshot
customerEmail	String	Y			Snapshot
customerAddress	String	Y			Snapshot
customerPartyGstNo	String	Y			Snapshot
customerPartyState	String	Y			Snapshot
discountPercent	Decimal(5,2)	N	0		Order discount % last applied
subtotal	Decimal(14,2)	N	0		Sum of qty × price
discountAmount	Decimal(14,2)	N	0		Sum of line discounts
taxableAmount	Decimal(14,2)	N	0		Subtotal – discount
cgstAmount	Decimal(14,2)	N	0		CGST
sgstAmount	Decimal(14,2)	N	0		SGST
igstAmount	Decimal(14,2)	N	0		IGST
taxAmount	Decimal(14,2)	N	0		Line tax sum
transportAmount	Decimal(14,2)	N	0		Freight after tax
totalAmount	Decimal(14,2)	N	0		Grand total
note	String	Y			Note
createdById	String	N		FK → User	Creator
convertedSaleId	String	Y		UK, FK → Sale	Future conversion hook
createdAt	DateTime	N	now()		Created
updatedAt	DateTime	N	auto		Updated

Indexes: unique quotationNumber, unique convertedSaleId, customerId, status, quotationDate, validUntil, createdById

4.14c QuotationLine

productId is the catalog product. Name, SKU, HSN, unit, and GST percents are snapshots.

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Line id

Field	Type	Null	Default	Keys	Description
quotationId	String	N		FK → Quotation	Parent (Cascade)
productId	String	N		FK → Product	Catalog product
productUnitId	String	Y		FK → ProductUnit	Unit (SetNull)
quantity	Decimal(18,4)	N			Qty in selected unit
quantityInBase	Decimal(18,4)	N			Qty × conversion
unitPrice	Decimal(14,4)	N			Selling price × conversion
lineDiscount	Decimal(14,2)	N	0		Discount
lineTax	Decimal(14,2)	N	0		GST amount
lineTotal	Decimal(14,2)	N			Taxable + tax
productName	String	N			Snapshot
sku	String	N			Snapshot
description	String	Y			Snapshot
hsnCode	String	Y			Snapshot
unitCode	String	N			Snapshot
unitDisplayName	String	N			Snapshot
cgstPercent	Decimal(5,2)	N	0		Rate snapshot
sgstPercent	Decimal(5,2)	N	0		Rate snapshot
igstPercent	Decimal(5,2)	N	0		Rate snapshot
createdAt	DateTime	N	now()		Created
updatedAt	DateTime	N	auto		Updated

Indexes: quotationId, productId, productUnitId

4.15 StockMovement

Ledger of quantity changes (base and unit qty).

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Movement id
type	StockMovementType	N			Movement kind
productId	String	N		FK → Product	Product
productUnitId	String	Y		FK → ProductUnit	Unit context

Field	Type	Null	Default	Keys	Description
quantity	Decimal(18,4)	N			Qty in unit (if any)
quantityInBase	Decimal(18,4)	N			Qty in base units
note	String	Y			Note
purchaseId	String	Y		FK → Purchase	Link when from purchase
saleId	String	Y		FK → Sale	Link when from sale
createdById	String	N		FK → User	Actor
createdAt	DateTime	N	now()		Created

Indexes: (productId, createdAt), (type, createdAt), purchaseId, saleId

4.16 StockAdjustment

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Adjustment id
productId	String	N		FK → Product	Product
quantityBefore	Decimal(18,4)	N			Stock before (base)
quantityAfter	Decimal(18,4)	N			Stock after (base)
difference	Decimal(18,4)	N			After – before
reason	String	N			Required reason
note	String	Y			Optional note
adjustedById	String	N		FK → User	Who adjusted
createdAt	DateTime	N	now()		Created

Indexes: (productId, createdAt), (adjustedById, createdAt)

4.17 ReportScheduleConfig

Singleton schedule (id defaults to "default").

Field	Type	Null	Default	Keys	Description
id	String	N	"default"	PK	Singleton key
notifyEmail	String	N	""		Owner notify email

Field	Type	Null	Default	Keys	Description
salesEnabled	Boolean	N	false		Enable sales schedule
salesPeriod	ReportPeriod	Y			Sales cadence
inventoryEnabled	Boolean	N	false		Enable inventory
inventoryPeriod	ReportPeriod	Y			Inventory cadence
supplierOutstandingEnabled	Boolean	N	false		Enable supplier OS
supplierOutstandingPeriod	ReportPeriod	Y			Supplier OS cadence
customerOutstandingEnabled	Boolean	N	false		Enable customer OS
customerOutstandingPeriod	ReportPeriod	Y			Customer OS cadence
updatedAt	DateTime	N	auto		Last config change

4.18 ReportRun

Field	Type	Null	Default	Keys	Description
id	String	N	cuid()	PK	Run id (used in ?reportRun= links)
reportType	ScheduledReportType	N			Which report
period	ReportPeriod	N			Cadence used
periodStart	DateTime	N		UK*	Period start (*with type+period)
periodEnd	DateTime	N			Period end
asOf	DateTime	N			Generation timestamp
status	ReportRunStatus	N			Run status
payload	Json	Y			Stored report body
errorMessage	String	Y			Failure detail
emailSentAt	DateTime	Y			When email sent
createdAt	DateTime	N	now()		Created

Unique: (reportType, period, periodStart)

Indexes: createdAt, (reportType, status)

5. Relationship summary (FK map)

From	Field	To	On delete (Prisma)
ProductUnit	productId	Product	Cascade
Barcode	productId	Product	Cascade
Barcode	productUnitId	ProductUnit	SetNull
PromotionProduct	promotionId	Promotion	Cascade
PromotionProduct	productId	Product	Cascade
Purchase	supplierId	Supplier	Restrict (default)
Purchase	createdById	User	Restrict
PurchaseLine	purchaseId	Purchase	Cascade
PurchaseLine	productId	Product	Restrict
PurchaseLine	productUnitId	ProductUnit	Restrict
PurchasePayment	purchaseId	Purchase	Cascade
PurchasePayment	createdById	User	Restrict
Sale	customerId	Customer	SetNull
Sale	createdById	User	Restrict
SaleLine	saleId	Sale	Cascade
SaleLine	productId	Product	Restrict
SaleLine	productUnitId	ProductUnit	Restrict
SalePayment	saleId	Sale	Cascade
SalePayment	createdById	User	Restrict
Quotation	customerId	Customer	SetNull
Quotation	createdById	User	Restrict
Quotation	convertedSaleId	Sale	SetNull
QuotationLine	quotationId	Quotation	Cascade
QuotationLine	productId	Product	Restrict
QuotationLine	productUnitId	ProductUnit	SetNull
StockMovement	productId	Product	Restrict
StockMovement	productUnitId	ProductUnit	Restrict
StockMovement	purchaseId	Purchase	Restrict
StockMovement	saleId	Sale	Restrict
StockMovement	createdById	User	Restrict
StockAdjustment	productId	Product	Restrict
StockAdjustment	adjustedById	User	Restrict

6. Maintenance

- Schema changes land in prisma/schema.prisma + migrations first.
- Update **this file** (fields, enums, FK map) and ER_DIAGRAMS.md (visual relationships) in the same change set.
- Regenerate Word/PDF:

```
powershell -NoProfile -ExecutionPolicy Bypass -File scripts/export-doc.ps1 -  
    DocName DATA_DICTIONARY -Title "Hardware Inventory - Data  
    Dictionary"  
powershell -NoProfile -ExecutionPolicy Bypass -File scripts/export-doc.ps1 -  
    DocName ER_DIAGRAMS -Title "Hardware Inventory - ER Diagrams"
```

*End of Data Dictionary. Behavioral use of these entities is in docs/FSD.md. Diagrams:
docs/ER_DIAGRAMS.md.*